Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2026-27

Class:	**TE-AI&DS**	**Semester:**	**V**
Course Code:	**2015119**	**Course Name:**	**Agile SoftwareDevelopment& DevOps Lab**

Name of Student:	**Chirag Raut**
Roll No. :	85
Experiment No.:	**03**
Title of the Experiment:	**To set up a Continuous Integration Pipeline using Jenkins/GitHub Actions.**
Date of Performance:	
Date of Submission:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty : Ms. Kshitija Gharat

Signature :

Date :

Aim: To set up and execute a Continuous Integration Pipeline using Jenkins.

Objectives:

After completing this experiment, the student will be able to:

1. To understand the concept of Continuous Integration (CI).
2. To understand the Jenkins automation server.
3. To create and configure Jenkins Freestyle Projects.
4. To execute Shell/Batch commands using Jenkins.
5. To compile and execute Java programs using Jenkins.
6. To perform parameterized builds in Jenkins.
7. To execute Python programs using Jenkins.

Tools Required:

• Hardware:

- Desktop/Laptop

• Software:

- Jenkins
- Java JDK
- Python
- Web Browser
- Windows Command Prompt / PowerShell

Theory:

Continuous Integration (CI) is a software development practice in which developers frequently integrate their code changes into a shared repository. Each integration is verified automatically by building and testing the application, which helps in detecting errors at an early stage.

Jenkins is an open-source automation server used to automate various stages of software development, such as building, testing, and deploying applications. Jenkins supports Continuous Integration by automatically executing predefined tasks whenever a build is triggered.

In Jenkins, a **Freestyle Project** is one of the simplest project types used to configure and execute build tasks. It can be used to run shell commands, batch commands, Java programs, Python programs, and other automation tasks.

A **Parameterized Build** allows the user to provide input values at runtime. These parameters can then be used by the build commands or programs during execution.

In this experiment, Jenkins is used to create Freestyle Projects and perform different tasks such as executing commands, running Java programs, providing runtime parameters, and executing Python programs. The Jenkins build results and program output can be verified using the **Console Output**.

This experiment demonstrates the basic working of Jenkins and its role in implementing Continuous Integration. The provided experiment material specifically covers CI, Jenkins automation, Freestyle Projects, Java execution, parameterized builds, and Python execution.

Procedure:

1. Open the Jenkins environment.

2. Create a new Jenkins Freestyle Project.

3. Perform Example 1:

- Create a project named **Example1**.
- Select **Freestyle Project**.
- Add a build step.
- Execute a simple Shell/Batch command.

- Save the project.
- Click **Build Now**.
- Check the output using **Console Output**.

4. Perform Example 2:

- Write a Java program.

- Test the Java program locally.
- Create a Jenkins Freestyle Project named **Example2**.
- Configure the build step to compile and execute the Java program.
- Build the project.
- Verify the output in the Jenkins Console Output.

5. Perform Example 3: Parameterized Build

- Create a Freestyle Project named **Example3**.
- Enable **This project is parameterized**.
- Add a String Parameter for the user's name.
- Add a Choice Parameter for the city.
- Write commands to accept and display the parameters.
- Select **Build with Parameters**.
- Enter the required values.
- Execute the build and check the output.

6. Perform Example 4: Java Program with Parameters

- Write a Java program to display a multiplication table.
- Accept the number as a command-line argument.
- Create a parameterized Jenkins project named **Example4**.
- Create a parameter named **num**.
- Compile and execute the Java program using the parameter.
- Build the project with a value for **num**. ● Check the Console Output.

7. Perform Example 5: Python Program

- Write and test a Python program locally.
- Create a Jenkins Freestyle Project named **Example5**.

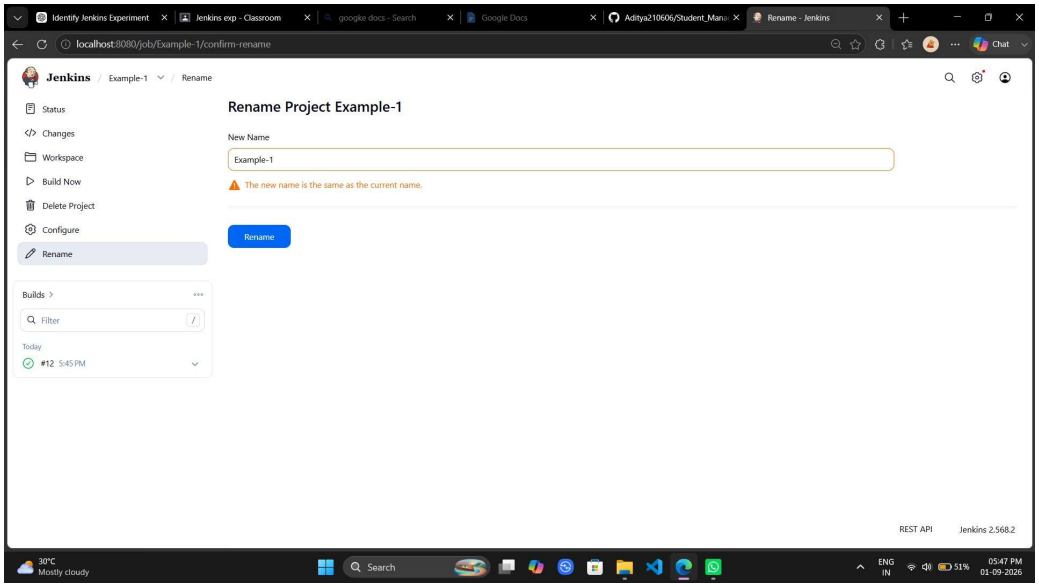
- Enable parameterized build.
- Create a parameter named **num**.
- Configure the build command to execute the Python program.
- Select **Build with Parameters**.
- Enter a value for **num**.
- Execute the build and verify the output.

Output: Example 1

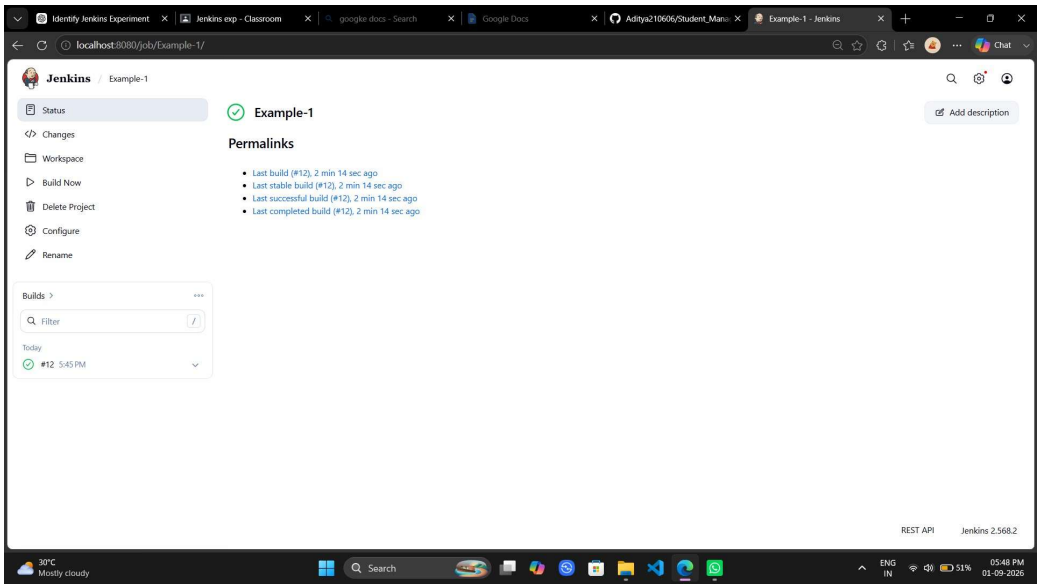
Lab 1.1: Deploying a Freestyle App in Jenkins

The Steps for deploying a simple free style project in Jenkins is as

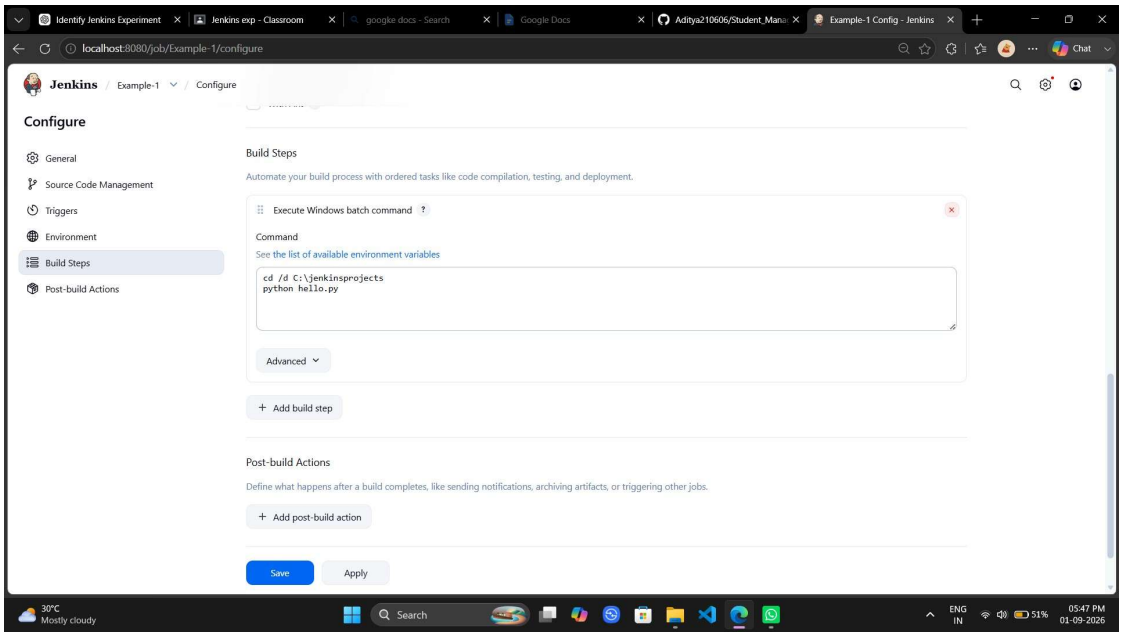
Follows. Step 1-: Click on Create new jobs.



Step 2-: Now Specify name to the project as "Example1", select Option "Free style project " and click on OK button



Step 3-: In this project we are going to learn how to run simple shell script on Jenkins. So, Click on Build option select Execute script from dropdown menu



Step 4-: Now Write a Simple Shell command to print the text as Like given below.

Command

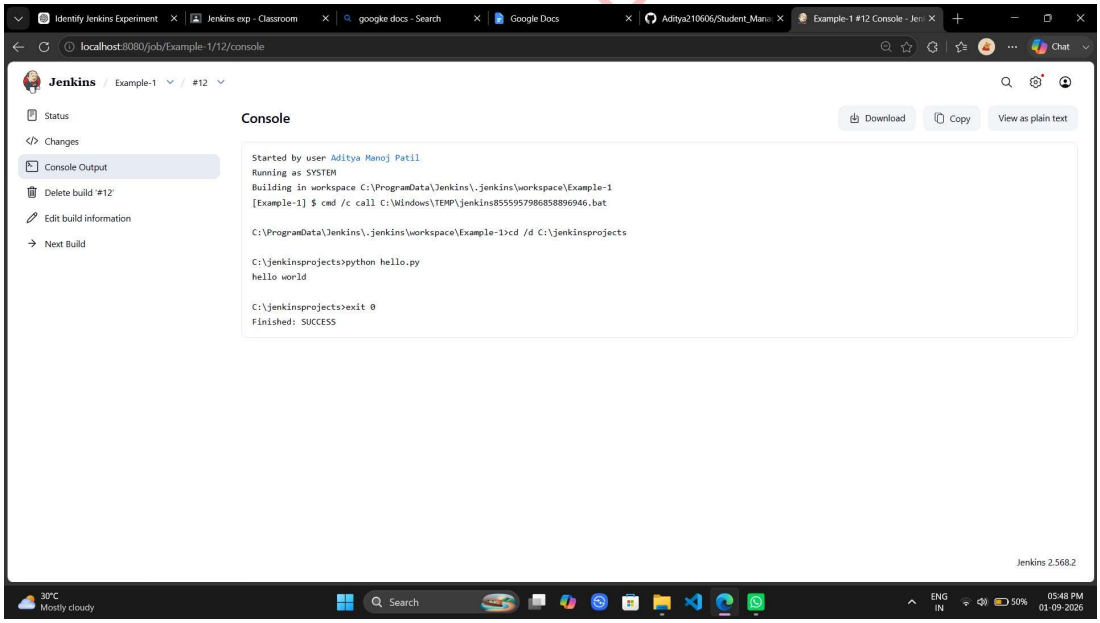
See the list of available environment variables

```
cd /d C:\jenkinsprojects
python hello.py
```

Advanced

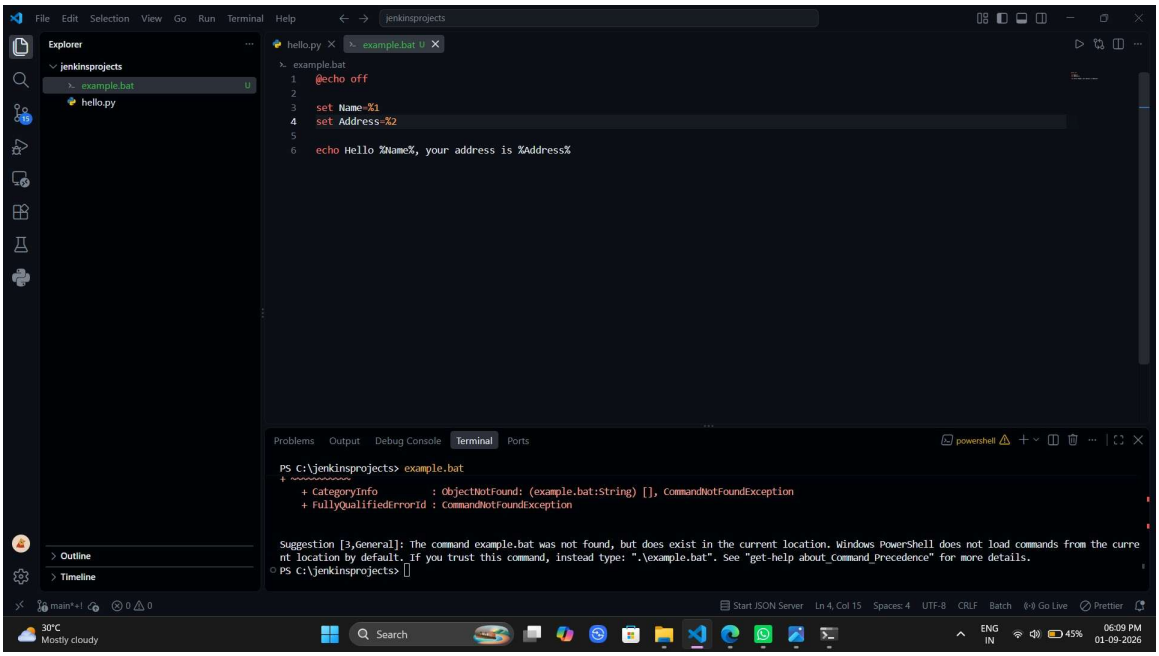
Bhushan Jadhav

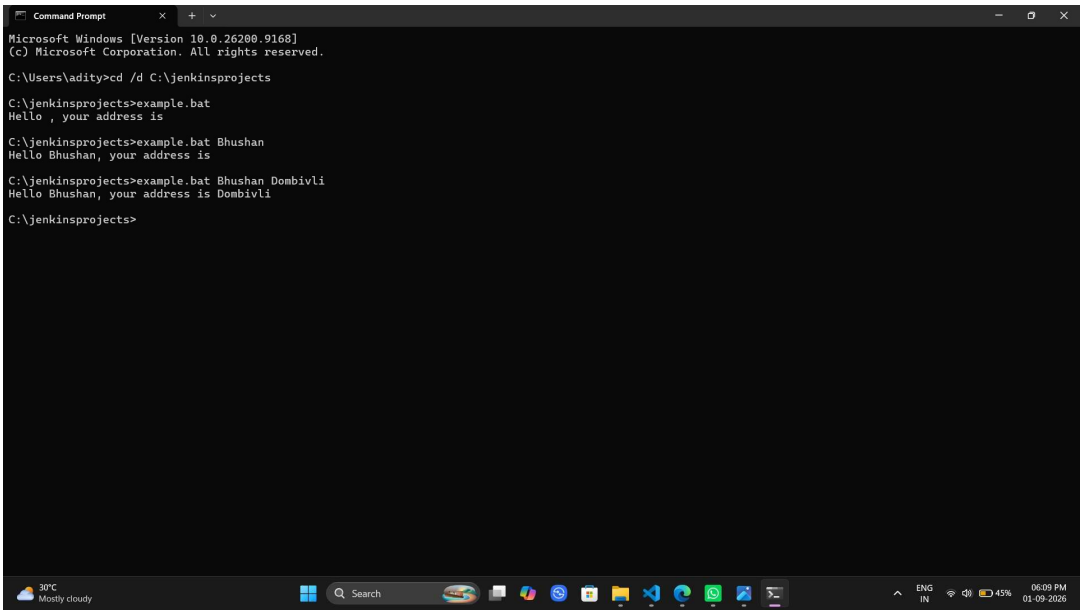
Step 5-: No Build a project to see the output Click on our first build "1" followed by console output to see the output

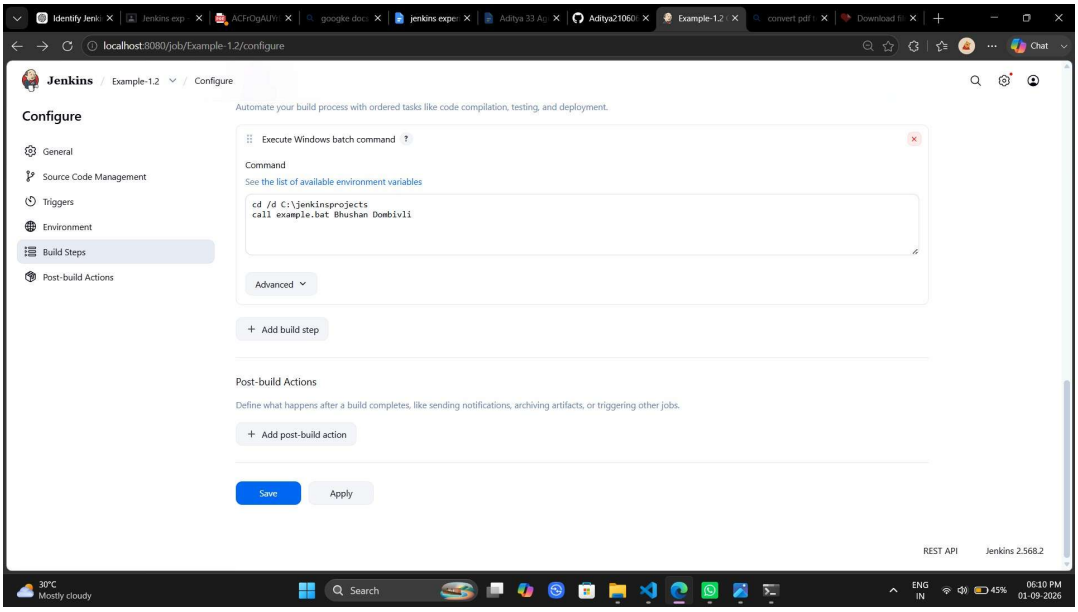


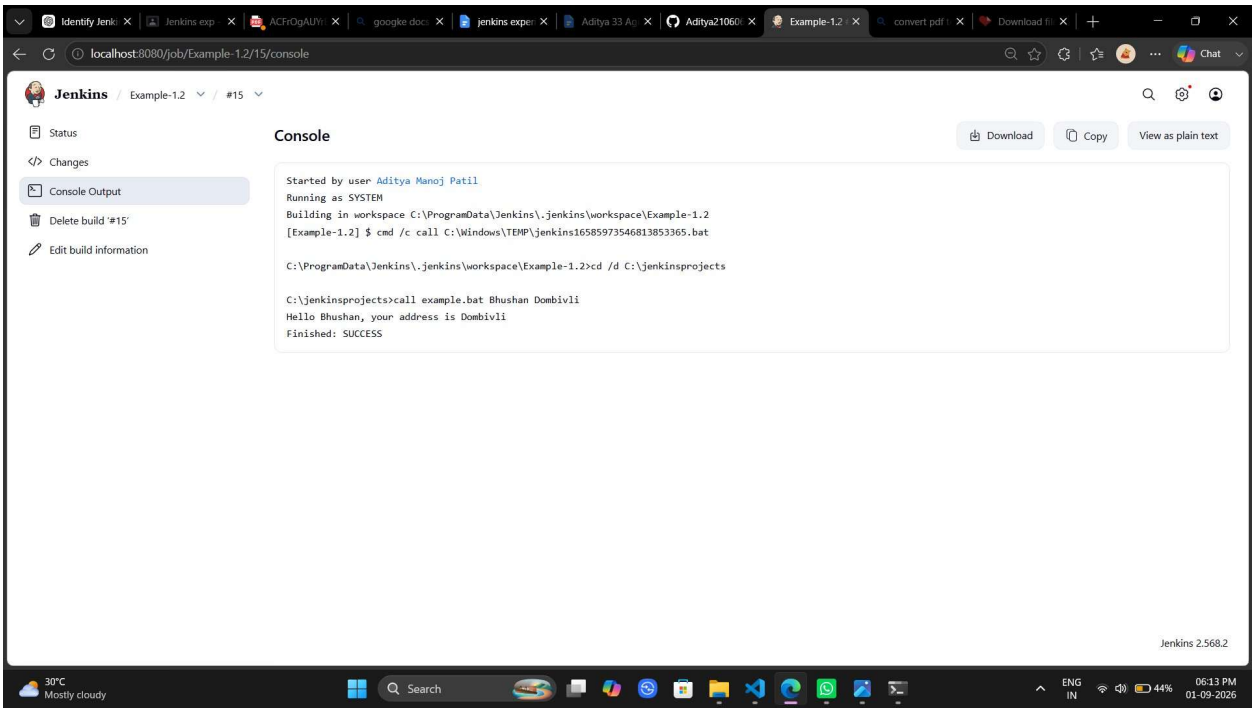
Example 1.2:

Now let us take parameters through files. So, create a new shell script file in local directory.

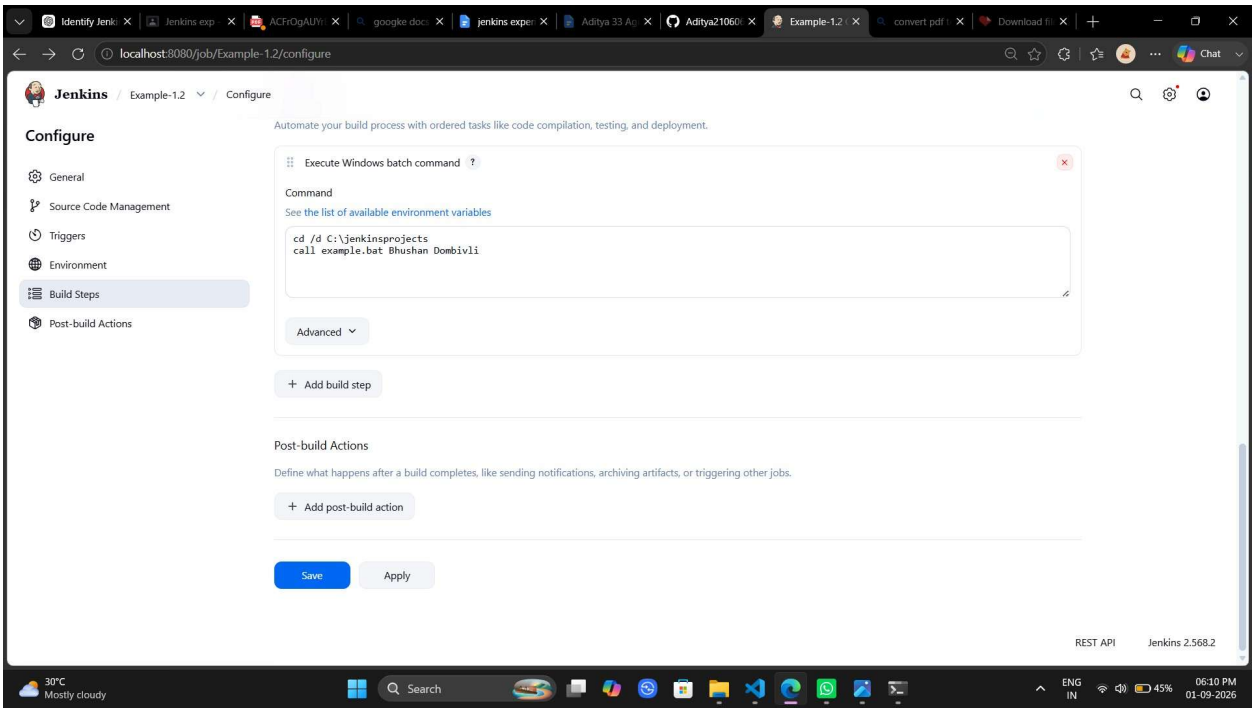

File Edit Selection View Go Run Terminal Help
jenkinsprojects
Explorer
jenkinsprojects
example.bat
hello.py
hello.py
example.bat
example.bat
@echo off
set Name=%1
set Address=%2
echo Hello %Name%, your address is %Address%
Problems Output Debug Console Terminal Ports
powershell
PS C:\jenkinsprojects> example.bat
+ CategoryInfo : ObjectNotFound: (example.bat:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException
Suggestion [3,General]: The command example.bat was not found, but does exist in the current location. Windows PowerShell does not load commands from the curre
nt location by default. If you trust this command, instead type: ".\example.bat". See "get-help about_Command_Precedence" for more details.
PS C:\jenkinsprojects>
Outline
Timeline
30°C
Mostly cloudy
Search
ENG
IN
45%
06:09 PM
01-09-2026

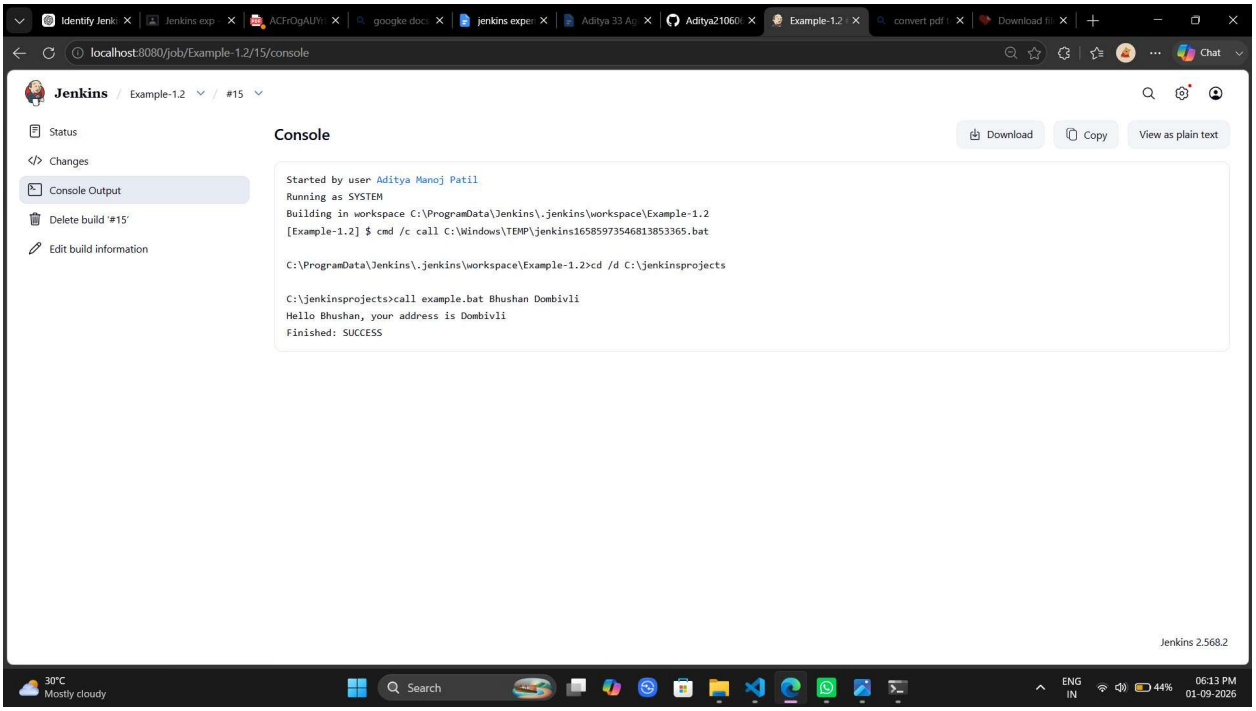

Command Prompt
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.
C:\Users\adity>cd /d C:\jenkinsprojects
C:\jenkinsprojects>example.bat
Hello , your address is
C:\jenkinsprojects>example.bat Bhushan
Hello Bhushan, your address is
C:\jenkinsprojects>example.bat Bhushan Dombivli
Hello Bhushan, your address is Dombivli
C:\jenkinsprojects>
30°C
Mostly cloudy
Search
ENG
IN
45%
06:09 PM
01-09-2026


localhost:8080/job/Example-1.2/configure
Jenkins
Example-1.2
Configure
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Automate your build process with ordered tasks like code compilation, testing, and deployment.
Execute Windows batch command
Command
See the list of available environment variables
cd /d C:\jenkinsprojects
call example.bat Bhushan Dombivli
Advanced
Add build step
Post-build Actions
Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.
Add post-build action
Save
Apply
REST API
Jenkins 2.568.2
30°C
Mostly cloudy
Search
ENG
IN
45%
06:10 PM
01-09-2026

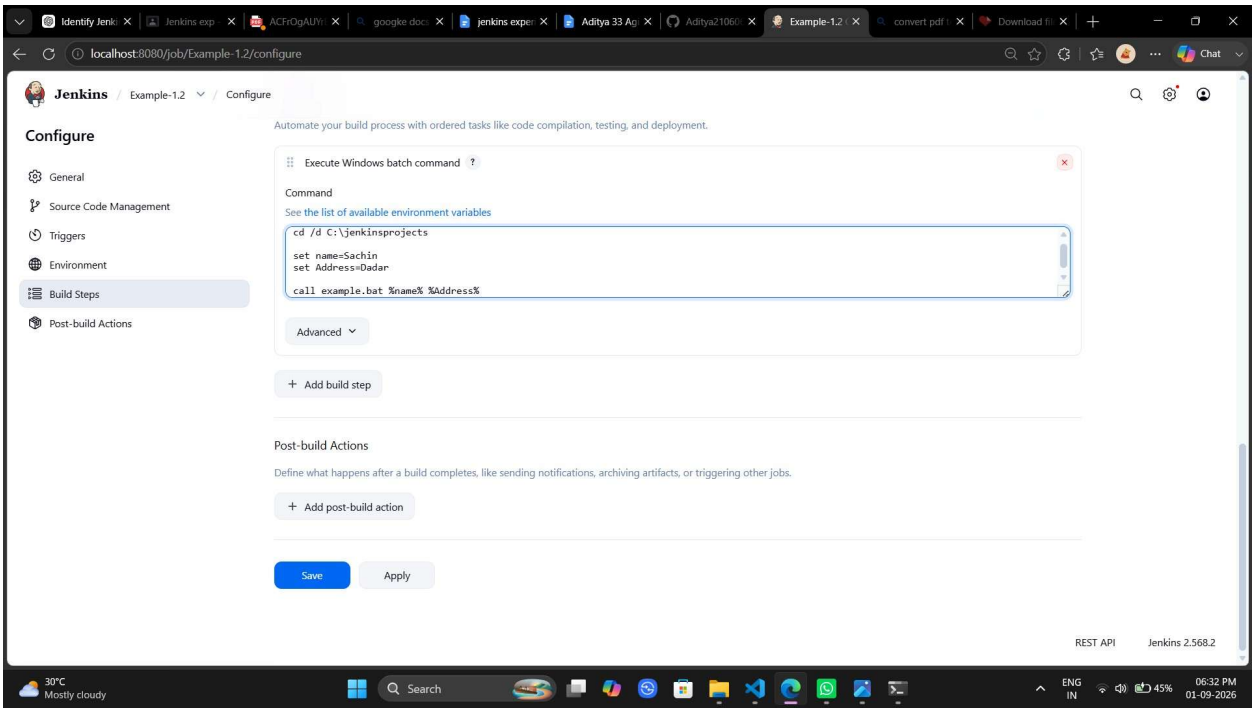


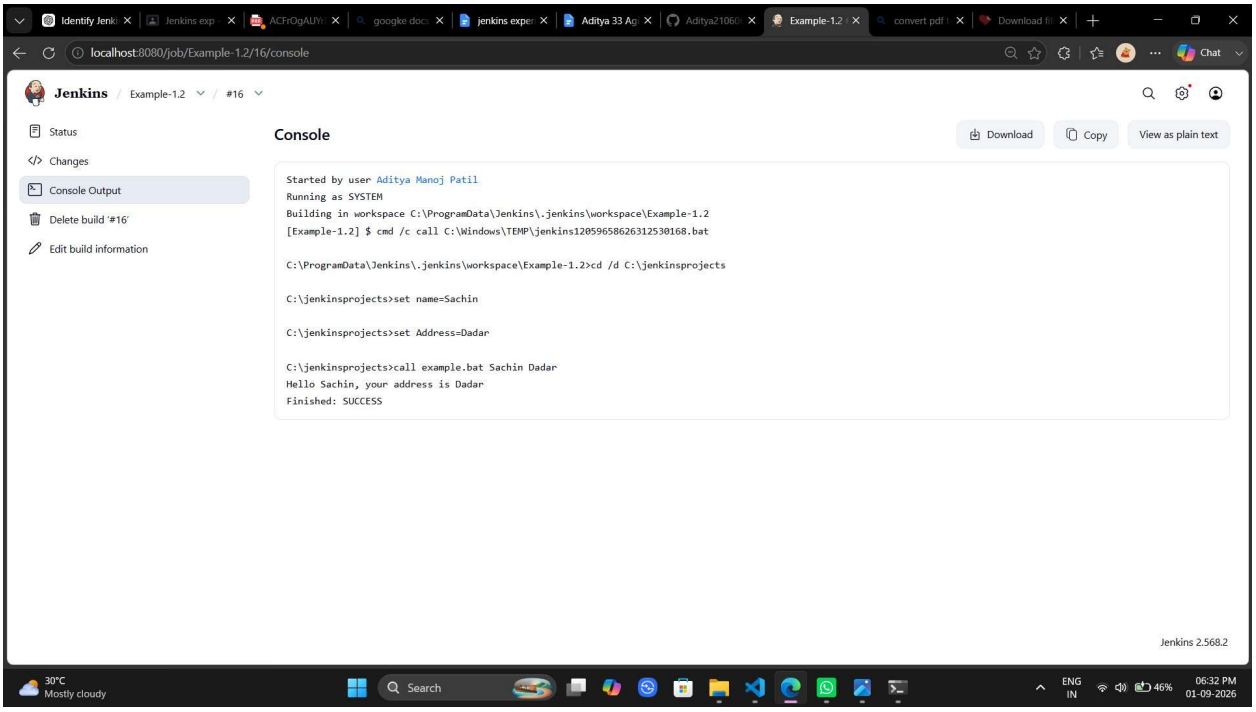
Now first run the shell script locally with no parameter, one parameter and two parameters. Now. Let us run it through Jenkins Shell. To change existing program, click on configure option and then modify the script. <Here, change directory to the path where you have stored your file..You can get the location by pwd command>


localhost:8080/job/Example-1.2/configure
Jenkins / Example-1.2 / Configure
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Automate your build process with ordered tasks like code compilation, testing, and deployment.
Execute Windows batch command
Command
See the list of available environment variables
cd /d C:\jenkinsprojects
call example.bat Bhushan Dombivli
Advanced
Add build step
Post-build Actions
Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.
Add post-build action
Save
Apply
REST API
Jenkins 2.568.2


localhost:8080/job/Example-1.2/15/console
Jenkins / Example-1.2 / #15
Status
Changes
Console Output
Delete build '#15'
Edit build information
Console
Download
Copy
View as plain text
Started by user Aditya Manoj Patil
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\.jenkins\workspace\Example-1.2
[Example-1.2] $ cmd /c call C:\Windows\TEMP\jenkins16585973546813853365.bat
C:\ProgramData\Jenkins\.jenkins\workspace\Example-1.2>cd /d C:\jenkinsprojects
C:\jenkinsprojects>call example.bat Bhushan Dombivli
Hello Bhushan, your address is Dombivli
Finished: SUCCESS
Jenkins 2.568.2

1.3 Variation to this program

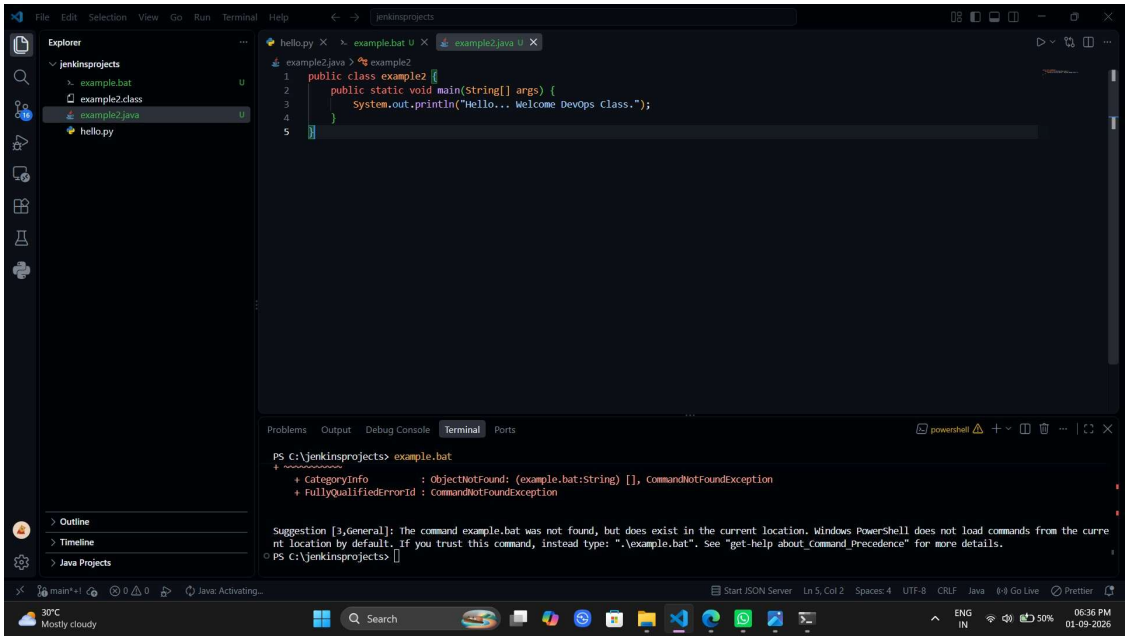




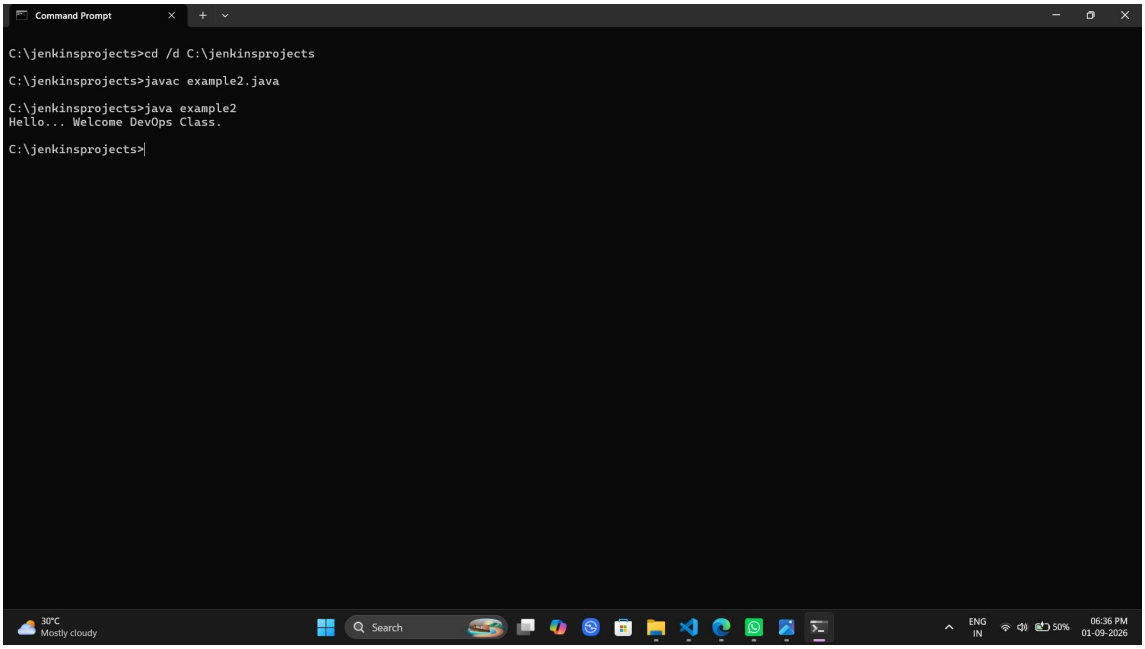
Example 2

1.4-: Running a Java Program under jenkins

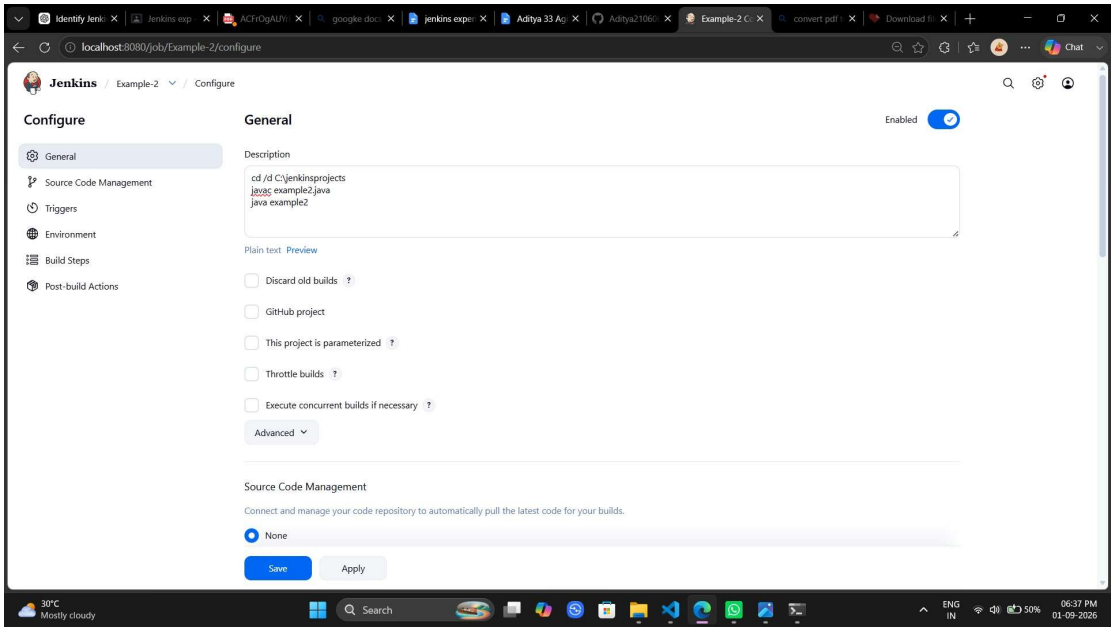
Step 1 -: Write a java program and test it locally



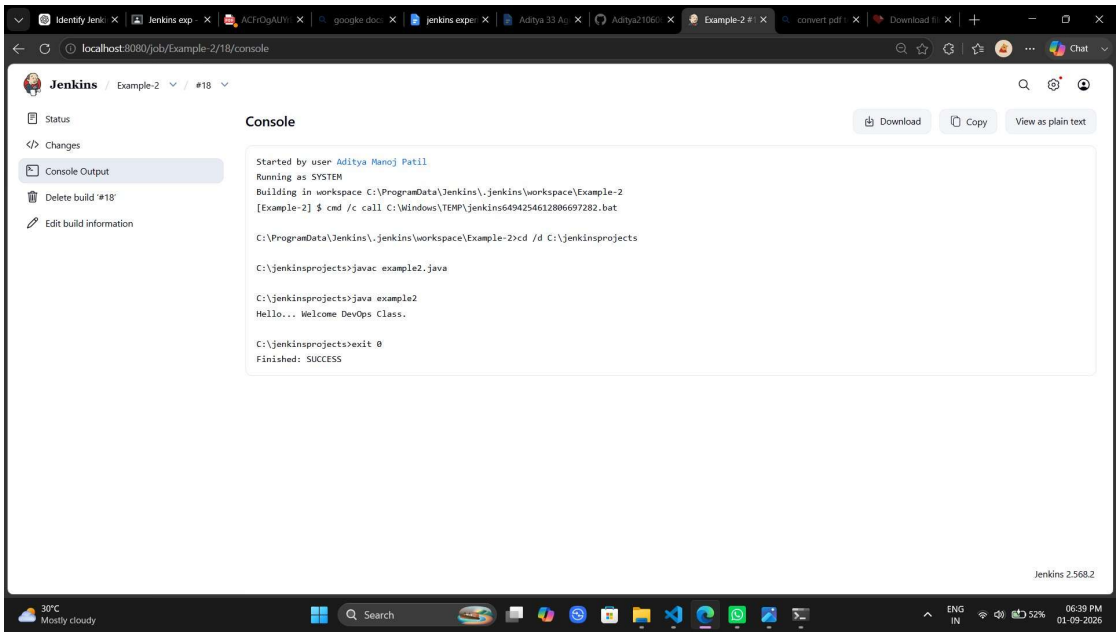
Testing :



Now create freestyle project example2 in jenkins. Go to build option and change path to directory where you have stored java file followed by compile and run program commands.



Now after doing this your build gets successful and get results

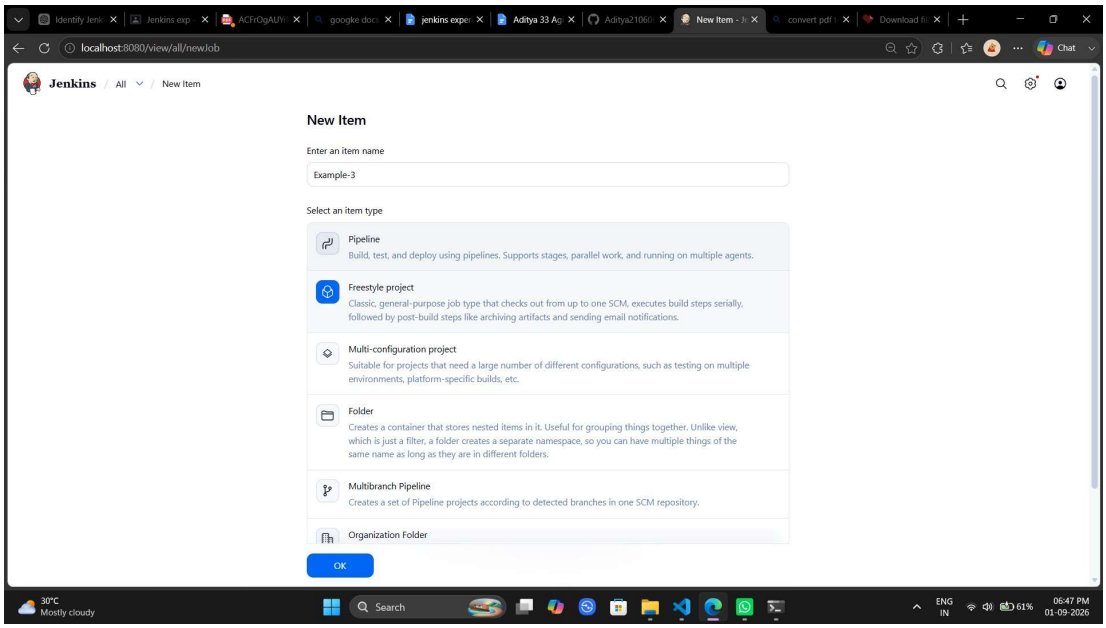


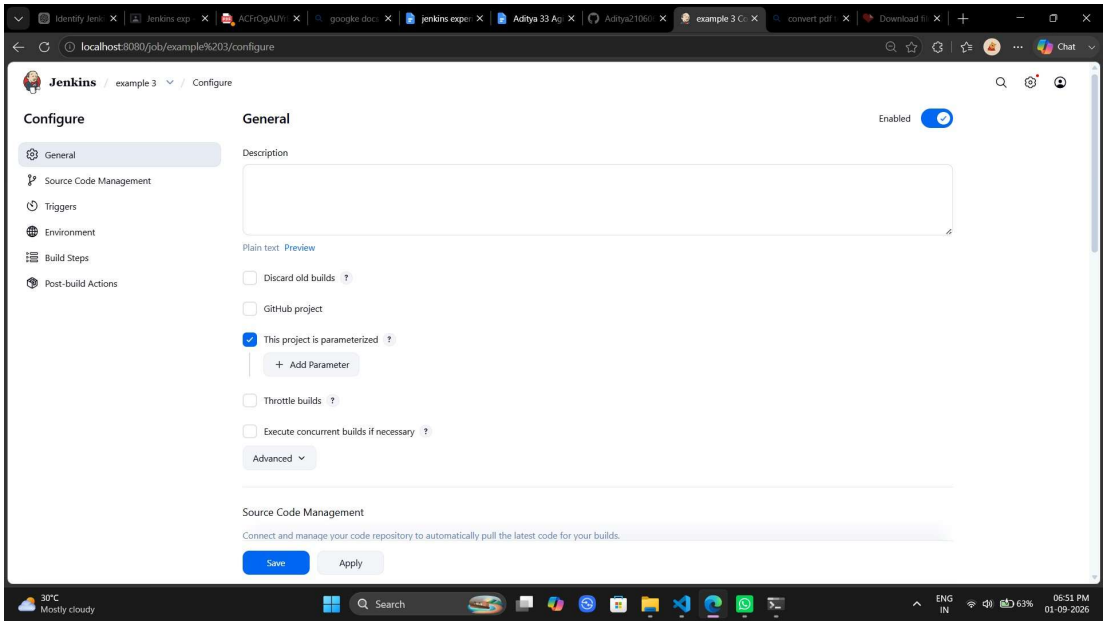
Example 3

Parametrize Build:

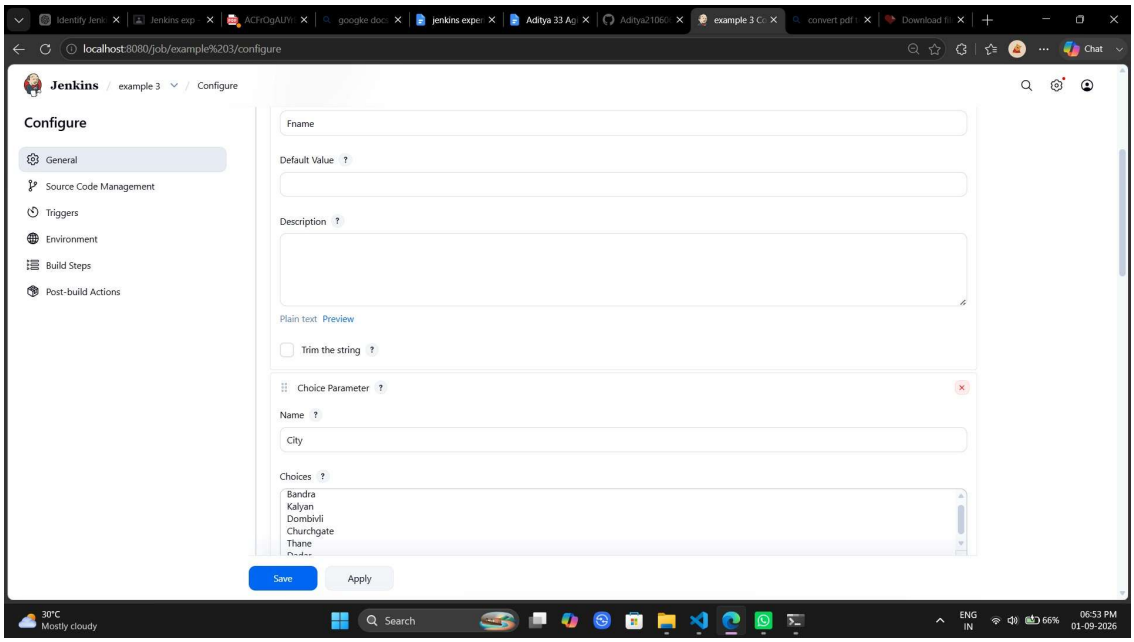
In this program we are going to see how to provide parameters during runtime to your shell script or java program.

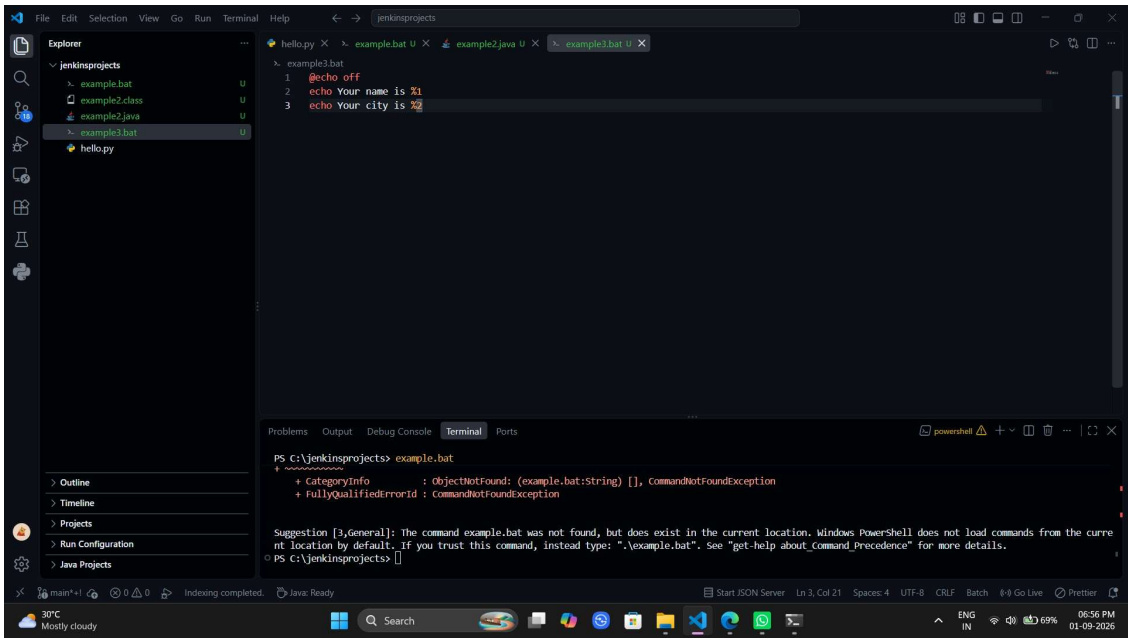
Step 1-: Create a free style project example 3 by clicking on new item followed by specifying project name and free style project.

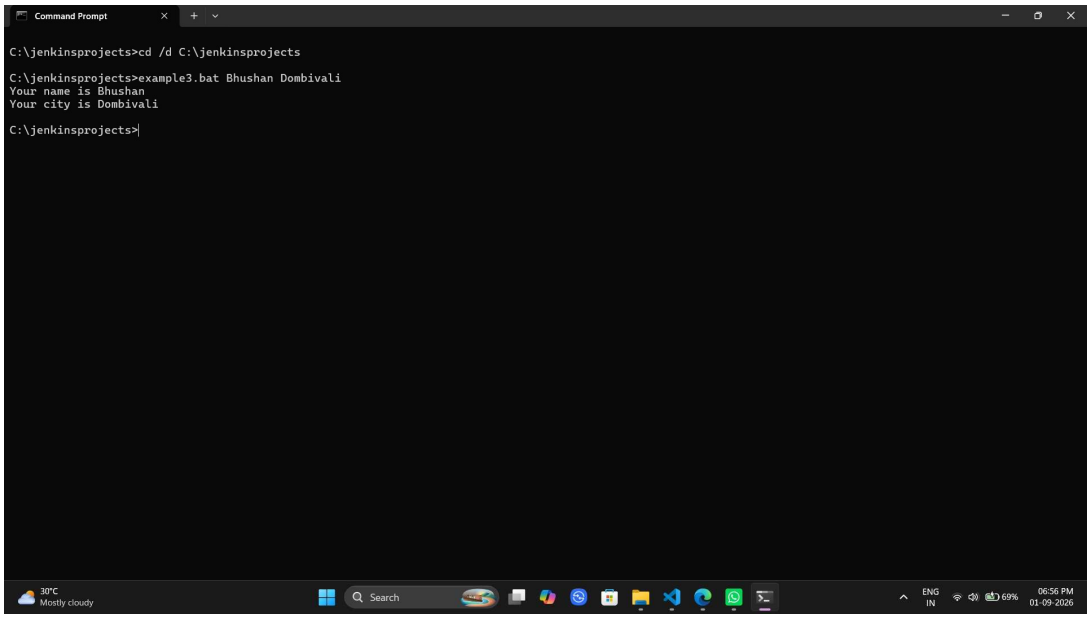




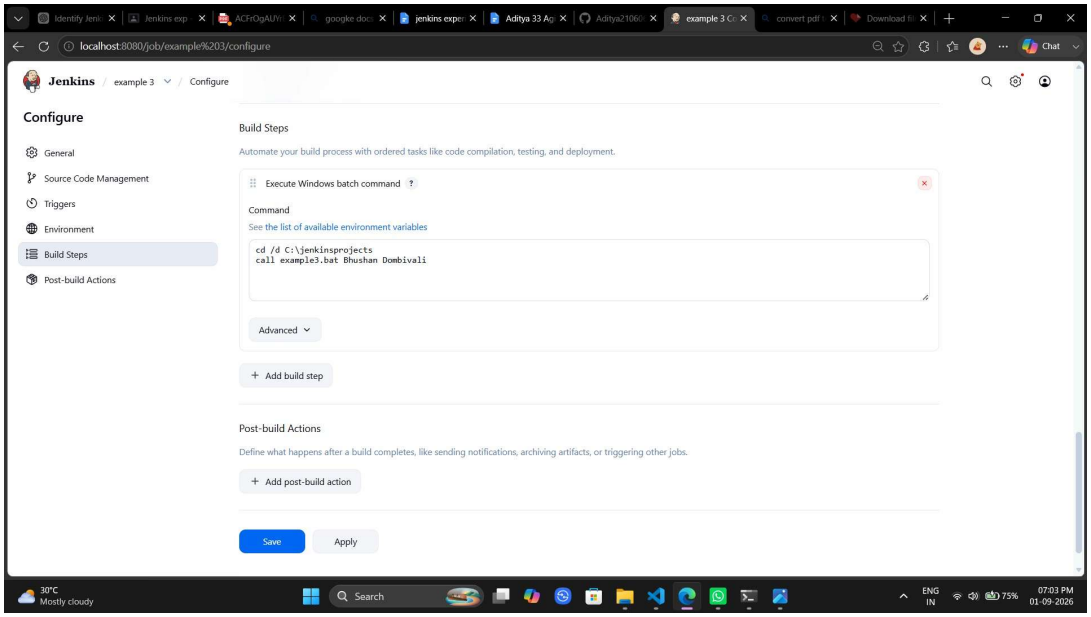
Again, click on add parameter and select choice parameter Take second parameter as choice box. Specify name as "City" and add the choices in each line


localhost:8080/job/example%203/configure
Jenkins
example 3
Configure
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Fname
Default Value
Description
Plain text
Preview
Trim the string
Choice Parameter
Name
City
Choices
Bandra
Kalyan
Dombivli
Churchgate
Thane
Save
Apply
30°C
Mostly cloudy
Search
ENG
IN
06:53 PM
01-09-2026

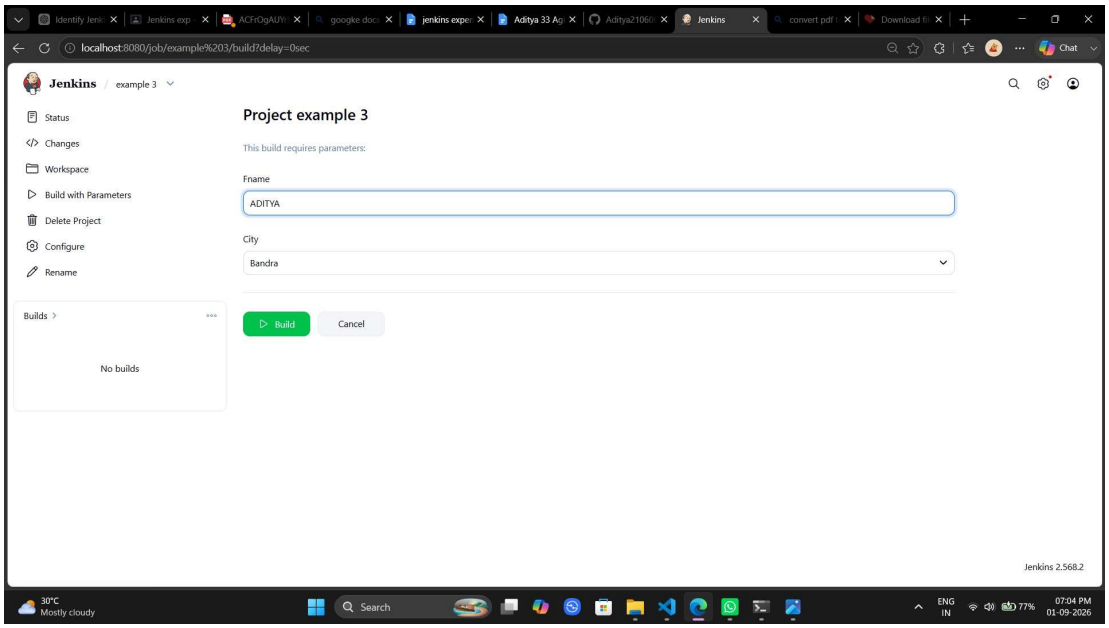

Explorer
jenkinsprojects
example.bat
example2.class
example2.java
example3.bat
hello.py
example3.bat
@echo off
echo Your name is %1
echo Your city is %2
Problems
Output
Debug Console
Terminal
Ports
PS C:\jenkinsprojects> example.bat
+ CategoryInfo : ObjectNotFound: (example.bat:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException
Suggestion [3,General]: The command example.bat was not found, but does exist in the current location. Windows PowerShell does not load commands from the current location by default. If you trust this command, instead type: ".\example.bat". See "get-help about_Command_Precedence" for more details.
PS C:\jenkinsprojects>
Outline
Timeline
Projects
Run Configuration
Java Projects
30°C
Mostly cloudy
Search
ENG
IN
06:56 PM
01-09-2026



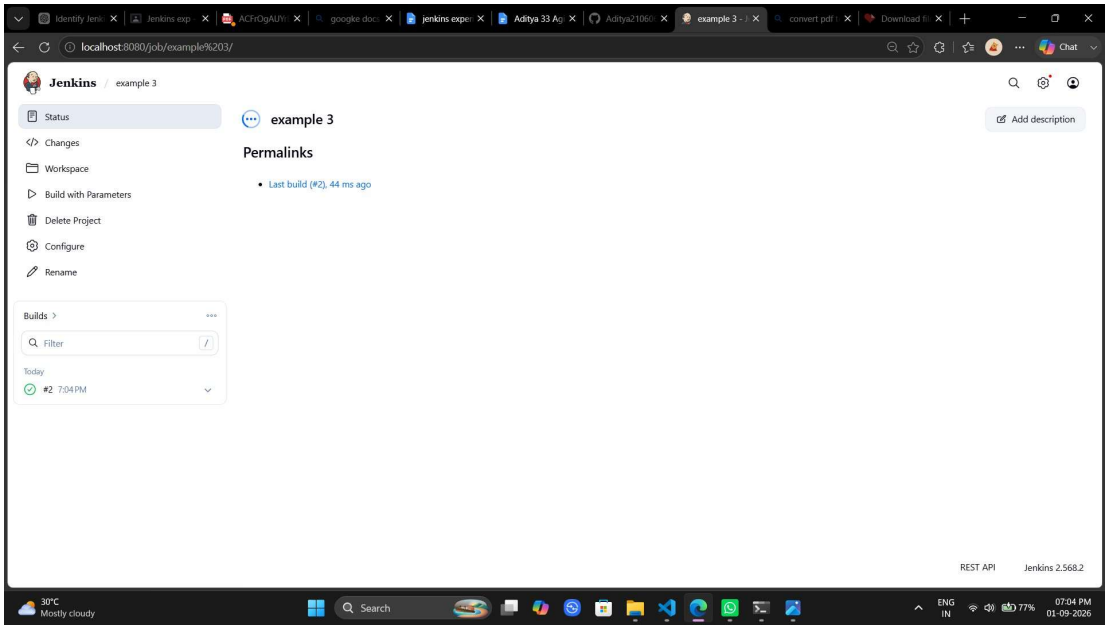
Now, go back to jenkins, Selct Build option, give the path and write script as shown below



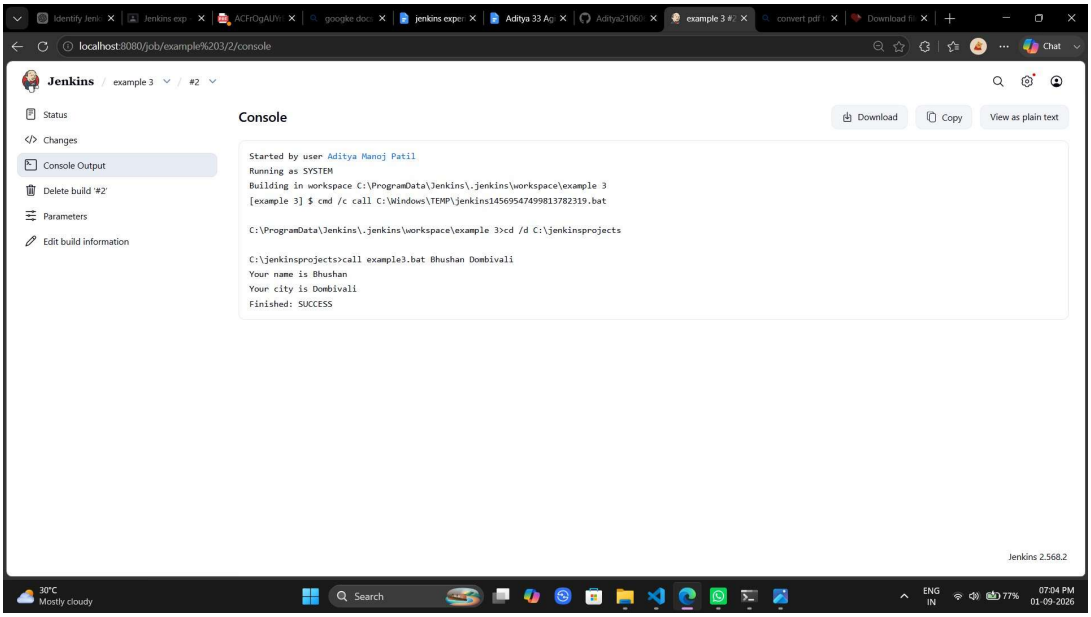
Now click on build with parameters and specify the values



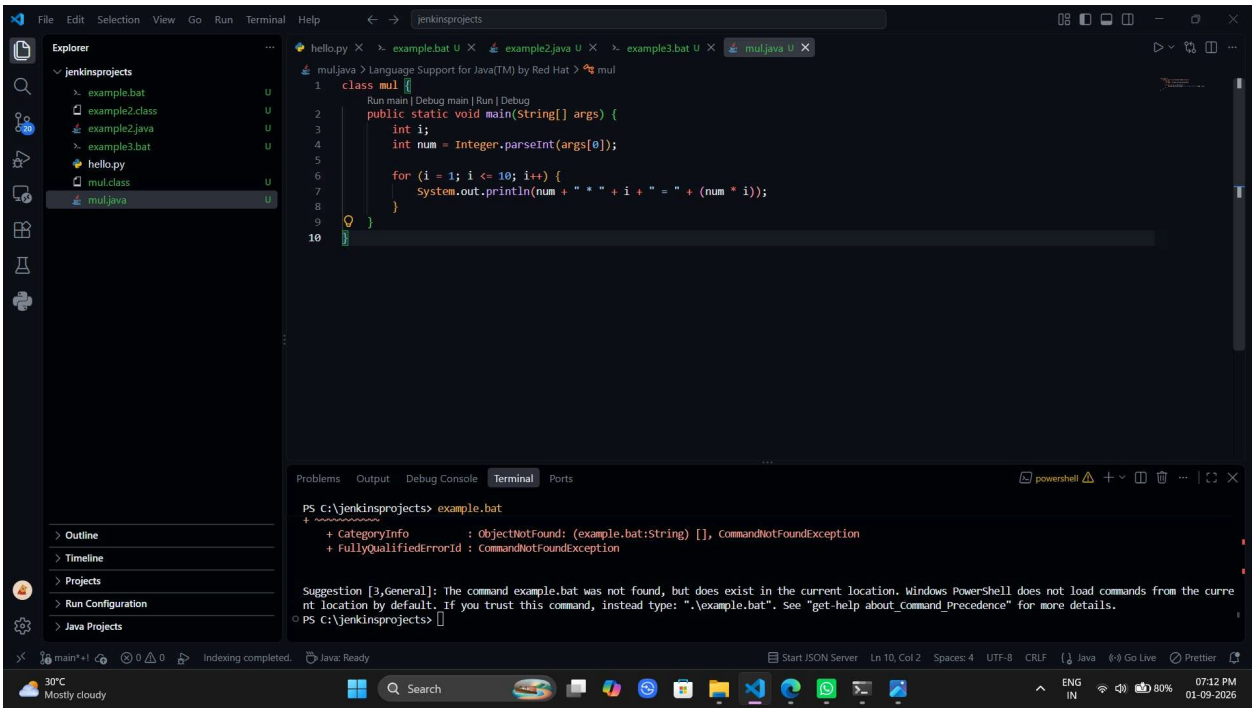
Click on Build



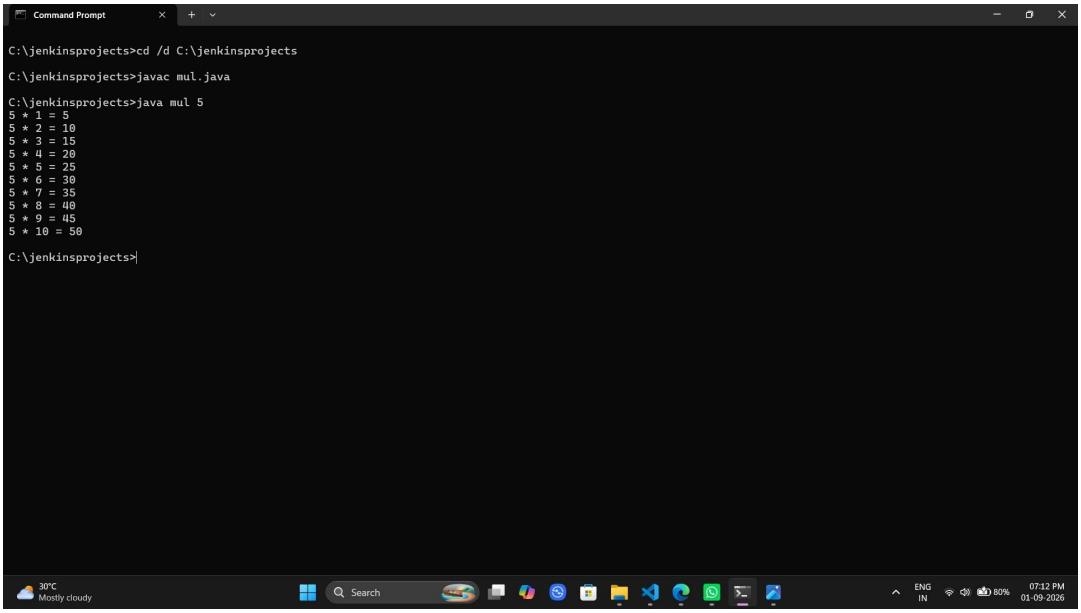
Go to console to see the output

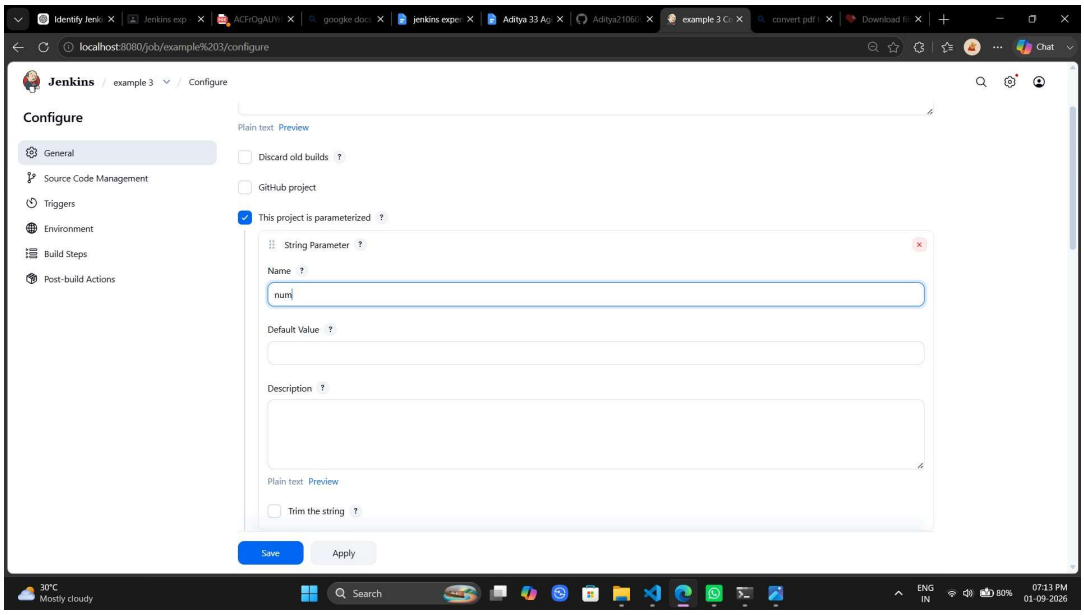


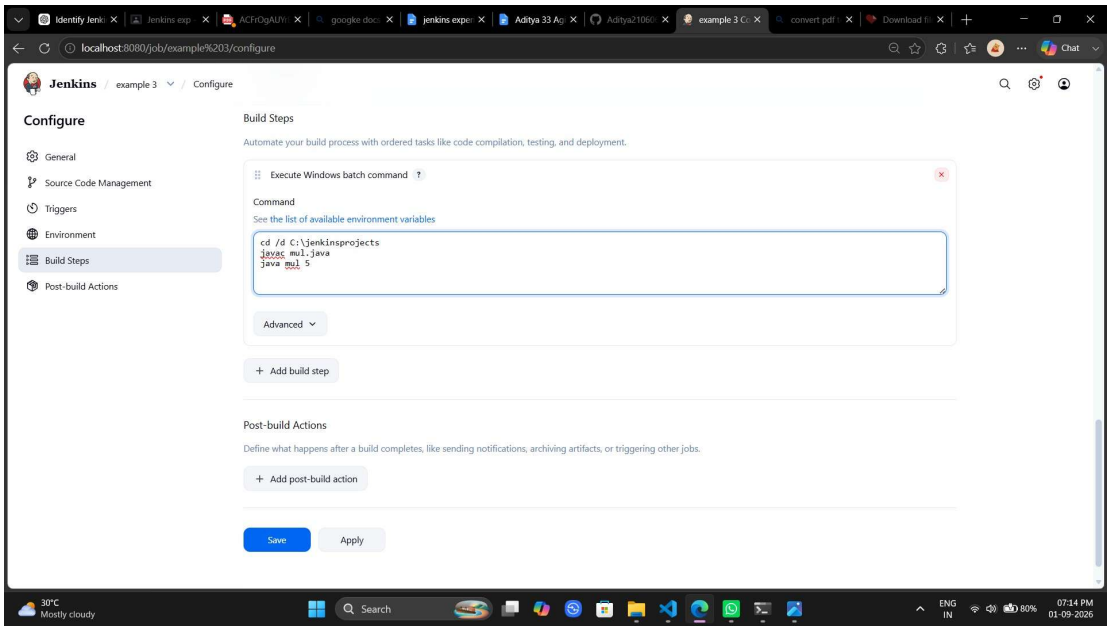
1.5 : Running a Java program with parameters :



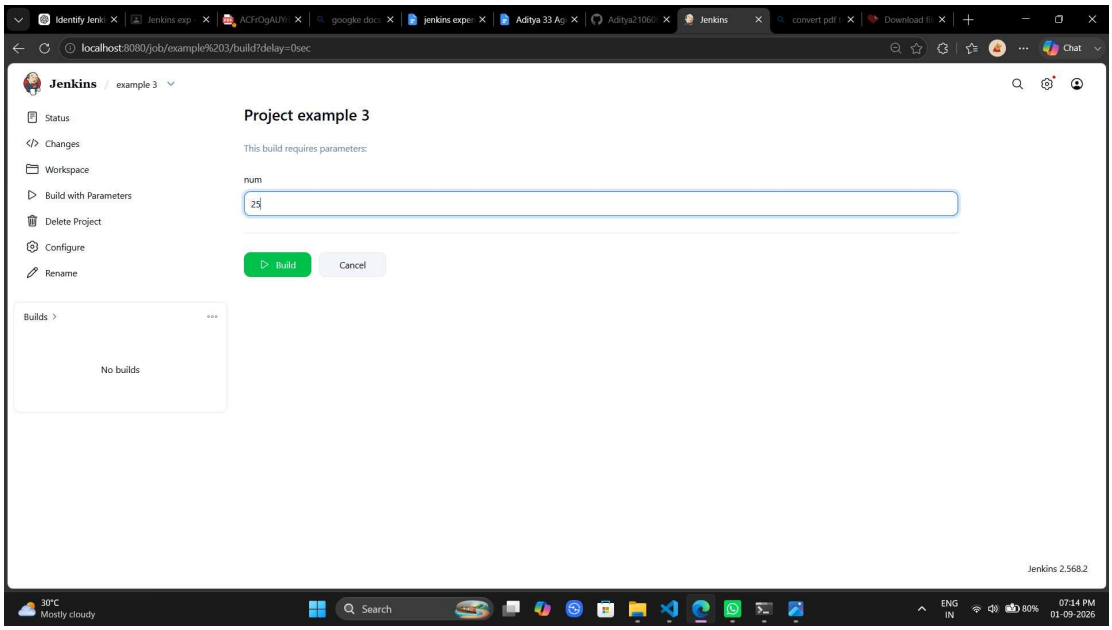
Write a java program for multiplication table with command line arguments and test it locally.



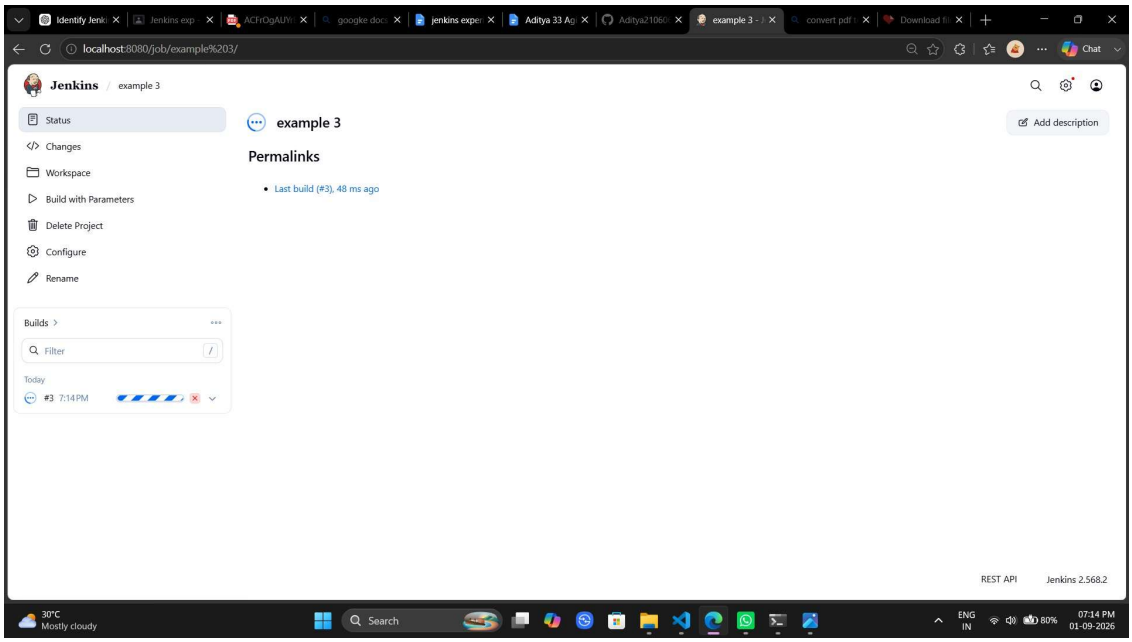

localhost:8080/job/example%203/configure
Jenkins
example 3
Configure
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Plain text Preview
Discard old builds
GitHub project
This project is parameterized
String Parameter
Name
num
Default Value
Description
Plain text Preview
Trim the string
Save
Apply
30°C
Mostly cloudy
Search
ENG
IN
80%
07:13 PM
01-09-2026



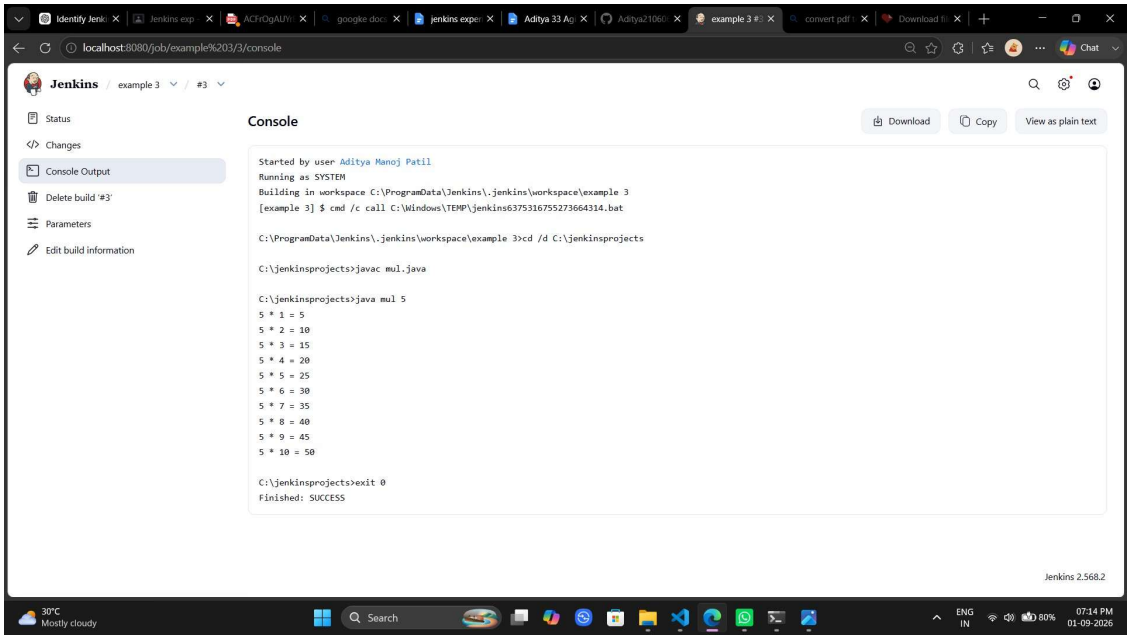
Now click on Save. Select build with parameter option and specify value of "num" whose multiplication tables needs to be displayed.



Build the project



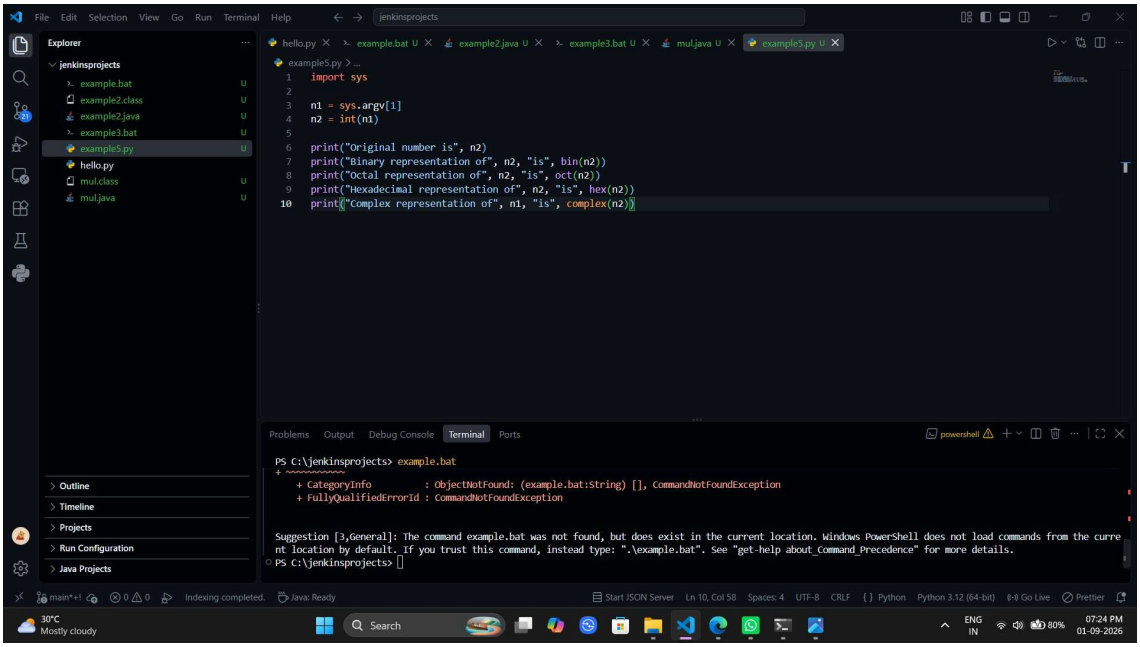
Click on build to see the output as below

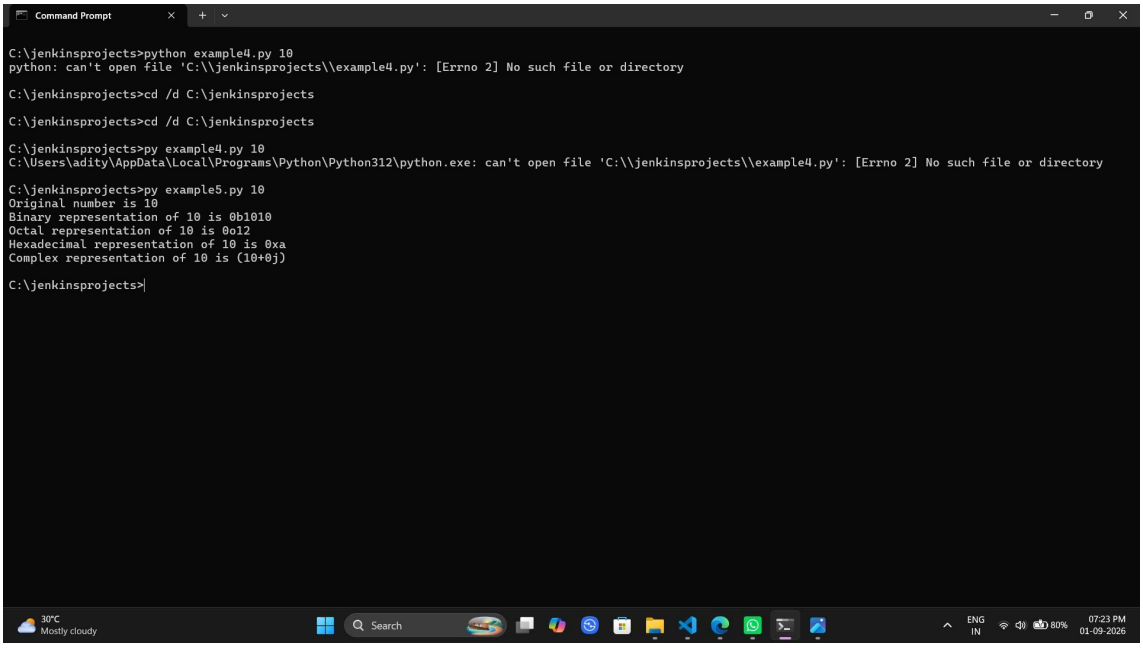


Example 5

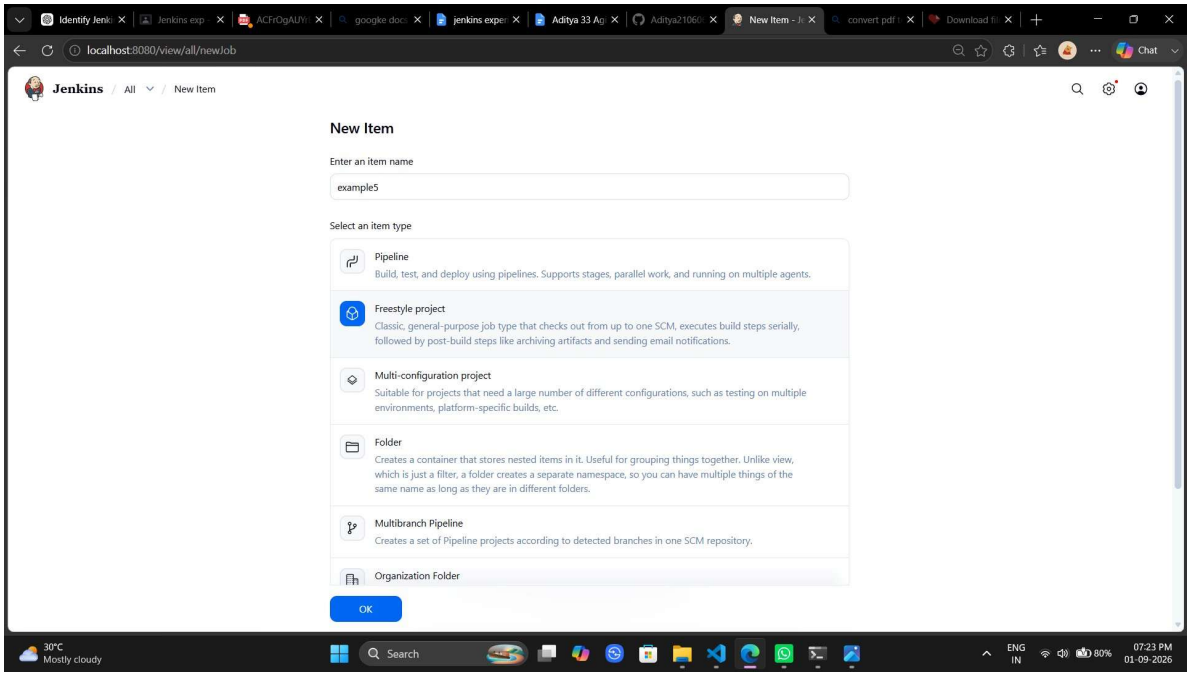
- **Running a python program in jenkins**

Step 1-: Write and test python3 program locally

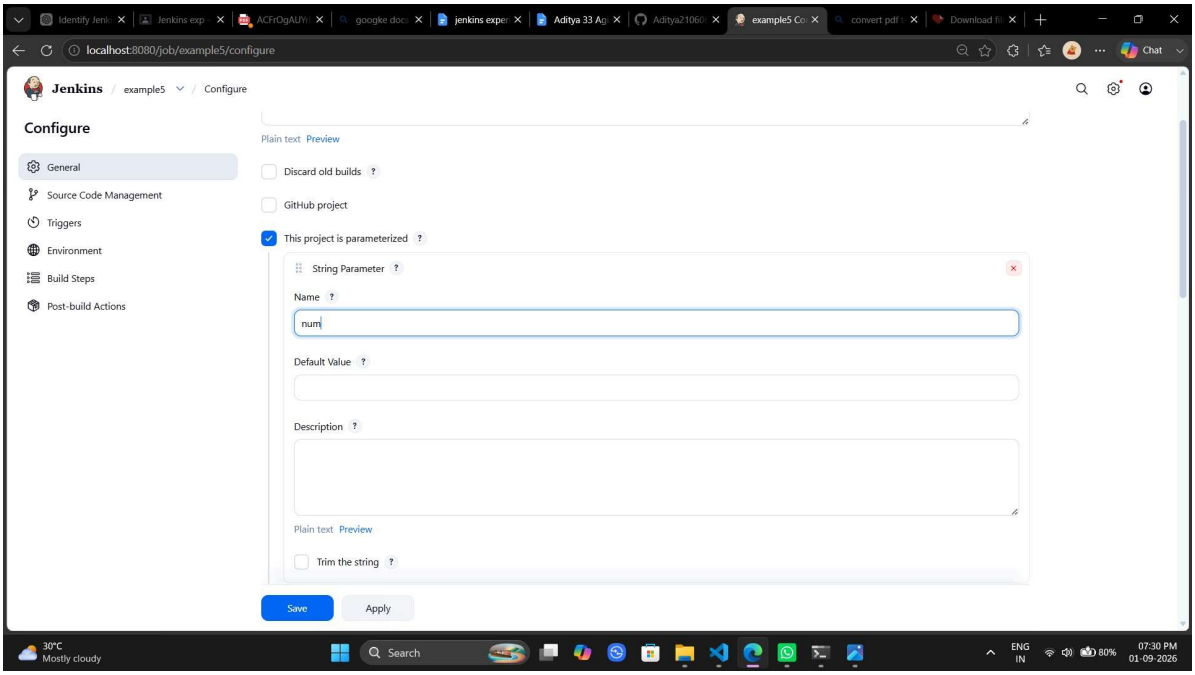




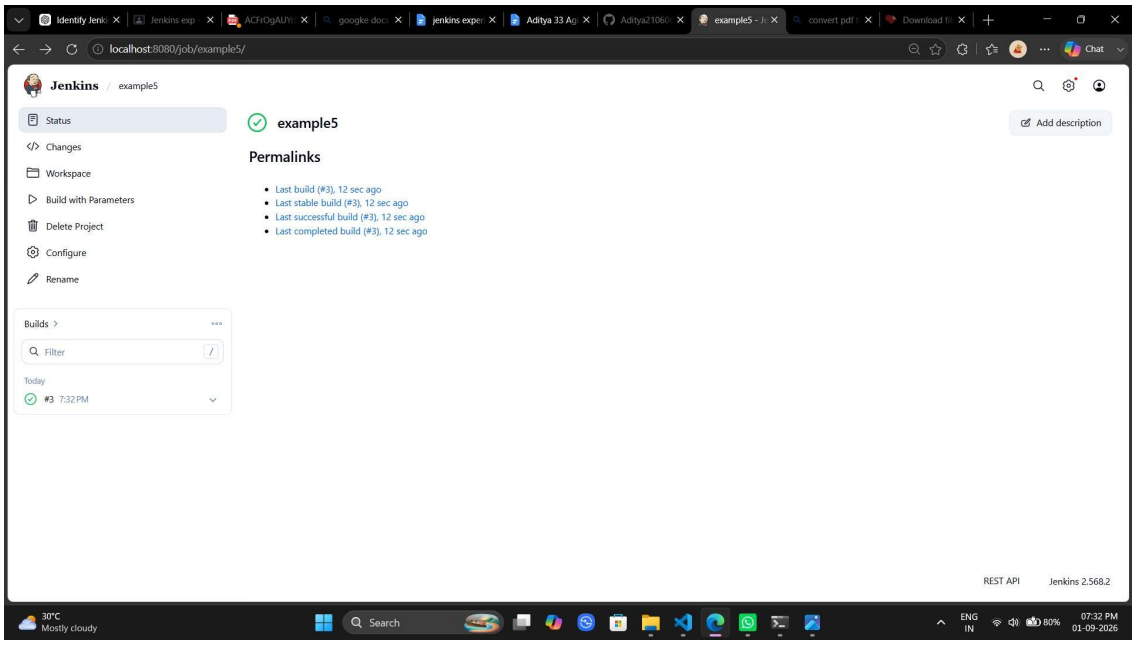
Now create a new item followed by specifying name "example5" and select freestyle project



Now select, this project is parameterized, select string parameter and specify parameter name as "num



Click on build and console to see the output :





Conclusion : This experiment demonstrated how Jenkins can automate the execution of programs on Windows. Java and Python programs were successfully executed using command-line arguments through Jenkins build steps. Thus, Jenkins helps automate the build and execution process efficiently.